Primes in Legendre intervals over $\mathbb{F}_q[t]$: an exact explicit-formula algorithm, verification for $d \le 10$, and an anomaly in characteristic 2

Ruqing Chen

GUT Geoservice Inc., Montréal, Canada

ruqing@hotmail.com

October 2026

DOI: 10.5281/zenodo.23225896 Code and data:
https://github.com/Ruqing1963/legendre-intervals-explicit-formula

Abstract

For monic $f \in \mathbb{F}_q[t]$ of degree d, the Legendre interval $\mathcal{I}_f = \{f^2 + s : \deg s \le d\}$ is the analogue of $[n^2, (n+1)^2]$. The Hayes–Weil estimate shows that $\mathcal{I}_f$ contains an irreducible polynomial whenever $q \ge d - 1$; the range $q \le d - 2$ is open. We give an algorithm that computes the prime count $\Psi(f) = \sum_{P \in \mathcal{I}_f} \Lambda(P)$ for *all* q^{d-1} intervals of a given (q, d) at once. It evaluates every Hayes L-function by a fast transform over the Hayes group, whose structure we make explicit for every $\mathbb{F}_q$, and runs in $O(d^2 q^{d-1} + d\,\sigma\, q^{d-1})$ operations, where σ is the sum of the orders of the cyclic factors in our decomposition of the group. Carried out in exact modular arithmetic, it proves that every Legendre interval contains an irreducible polynomial for every prime power q when $d \le 10$, and for $q = 2$ with $d \le 25$, $q = 3$ with $d \le 16$, $q = 4$ with $d \le 13$ and $q = 5$ with $d \le 11$. For odd q the fluctuations of Ψ across intervals follow the Keating–Rudnick variance $(d-2)q^{d+1}$ and a Gaussian law already for $q = 3$, and the form factor of the L-functions matches the circular unitary ensemble up to an explicit arithmetic diagonal term. In characteristic 2 the Legendre intervals form the subgroup of squares, every element has the shape $h^2 + tb^2$, and the variance of Ψ over them differs from the full-group value, which does follow $d-2$, by factors between 0.33 and 21. We show that this discrepancy is carried entirely by correlations between L-functions twisted by quadratic characters, and state its explanation as an open problem.

Contents

1 Introduction 2

2 Setup 2

3 The structure of the Hayes group 3

4 The algorithm 3

5 Verification 4

6 Fluctuations across intervals 5

7 The zeros 5

8 Characteristic 2 **7**
8.1 Structure 7
8.2 The anomaly 9

9 Discussion **11**

1 Introduction

Legendre's conjecture asserts that there is a prime between consecutive squares. Its function field analogue concerns the *Legendre interval*

$$\mathcal{I}_f = \{f^2 + s : \ s \in \mathbb{F}_q[t], \ \deg s \le d\}, \qquad f \in \mathbb{F}_q[t] \text{ monic of degree } d,$$

the set of monic polynomials of degree $2d$ whose coefficients of $t^{2d-1}, \dots, t^{d+1}$ agree with those of f^2. Write

$$\Psi(f) = \sum_{P \in \mathcal{I}_f} \Lambda(P), \qquad N_{\mathrm{irr}}(f) = \#\{P \in \mathcal{I}_f : \ P \text{ irreducible}\},$$

so that $2d\, N_{\mathrm{irr}}(f) = \Psi(f) - E(f)$ with $E(f)$ the contribution of proper prime powers.

What is known. Irreducible polynomials with prescribed leading coefficients have been studied since Hayes [10, 11, 18, 7, 14, 9, 8]. Hayes characters and Weil's theorem give $|\Psi(f) - q^{d+1}| \le (d-2)q^d$ in every characteristic (Proposition 5.2 below); with a bound for $E(f)$ (Lemma 5.1) this shows $N_{\mathrm{irr}}(f) \ge 1$ whenever $q \ge d-1$ [6]. At $q = d-2$ the main term and the error are of the same size, and for $q \le d-2$, which contains the genuine analogue (q fixed, $d \to \infty$), no method is known: the large-q methods for primes in short intervals and arithmetic progressions, Chebotarev's theorem with a big monodromy group [1] and the equidistribution of Frobenius for the characters of the Hayes group [13, 12] (see Rudnick's survey [16]), do not apply when q is fixed. Earlier computations [6] tested every element of every interval and reached $d \le 8$ for all q.

This paper.

(1) *An exact all-intervals algorithm* (Sections 3–4). The values $\Psi(f)$ for all q^{d-1} classes are obtained from the explicit formula, computing every Hayes L-function by a transform over the Hayes group. We determine an explicit basis of this group for every $\mathbb{F}_q$. Done modulo two primes and recombined, the computation is exact.

(2) *Verification* (Section 5). Every Legendre interval contains an irreducible polynomial for all q when $d \le 10$, and much further for small q.

(3) *Fluctuations and zeros* (Sections 6–7). For odd q, the variance across intervals agrees with Keating–Rudnick [13] at fixed q; the form factor of the L-functions equals an exact arithmetic diagonal for $n \le d-1$ and the circular unitary ensemble (CUE) value beyond.

(4) *Characteristic 2* (Section 8). The Legendre intervals are the squares of the Hayes group; the variance of their prime counts departs from the prediction by factors between 0.33 and 21. We show that the departure is carried by correlations between quadratic twists and pose its explanation as an open problem.

2 Setup

Let $\ell = d-1$. For monic g of degree k put $[g] = u^k g(1/u) \bmod u^{\ell+1}$, an element of

$$\mathcal{G} = \mathcal{G}_\ell = \{1 + c_1 u + \cdots + c_\ell u^\ell\} \quad \text{(multiplication mod } u^{\ell+1}), \qquad |\mathcal{G}| = q^\ell.$$

The map $g \mapsto [g]$ is multiplicative and records the ℓ coefficients below the leading one. For $x \in \mathcal{G}$ let $\Psi(x) = \sum_{\deg P = 2d,\, [P]=x} \Lambda(P)$; then $\Psi(f) = \Psi([f^2]) = \Psi([f]^2)$.

Lemma 2.1. *The interval $\mathcal{I}_f$ depends only on $f_{d-1}, \dots, f_1$; in particular $\mathcal{I}_{f+c} = \mathcal{I}_f$ for $c \in \mathbb{F}_q$. The map $f \mapsto [f]$ from monic polynomials of degree d to $\mathcal{G}$ is surjective and q-to-one, and $\mathcal{I}_f$ is the class $[f]^2$. Hence the Legendre intervals are exactly the classes in $\mathcal{G}^2 = \{x^2 : x \in \mathcal{G}\}$, distinct intervals being distinct classes. If q is odd, $\mathcal{G}^2 = \mathcal{G}$.*

Proof. For $c \in \mathbb{F}_q$, $(f+c)^2 - f^2 = 2cf + c^2$ has degree $\le d$, so $\mathcal{I}_{f+c} = \mathcal{I}_f$. The class $[f]$ records exactly $f_{d-1}, \dots, f_1$, which can be chosen freely; this gives surjectivity and the fibres of size q (the choice of f_0). Since $[\cdot]$ is multiplicative, $[f^2] = [f]^2$, and $\mathcal{I}_f$ is by definition the set of monic P of degree $2d$ with $[P] = [f^2]$. Squaring is a bijection on the finite p-group $\mathcal{G}$ when p is odd. □

For a character χ of $\mathcal{G}$ and $n \ge 1$ put $\psi_n(\chi) = \sum_{\deg g = n} \Lambda(g)\chi([g])$ and $c_k(\chi) = \sum_{\deg g = k} \chi([g])$. For $\chi \ne 1$, $L(z,\chi) = \sum_k c_k(\chi) z^k$ is a polynomial of degree $\le \ell - 1$ whose inverse roots have absolute value $q^{1/2}$ (Proposition 7.1 and [10, 13]), and $z\,L'(z,\chi)/L(z,\chi) = \sum_n \psi_n(\chi) z^n$. Comparing coefficients gives Newton's recursion

$$\psi_n(\chi) = n\,c_n(\chi) - \sum_{k=1}^{n-1} \psi_k(\chi)\, c_{n-k}(\chi), \qquad c_n = 0 \ (n \ge \ell), \tag{1}$$

which involves no division. By orthogonality, for every $x \in \mathcal{G}$,

$$\Psi(x) = q^{-\ell} \sum_{\chi} \overline{\chi(x)}\, \psi_{2d}(\chi), \qquad \psi_{2d}(1) = q^{2d}. \tag{2}$$

3 The structure of the Hayes group

Let $q = p^k$ and let $\omega_0, \dots, \omega_{k-1}$ be a basis of $\mathbb{F}_q$ over $\mathbb{F}_p$. For $j \ge 1$ with $p \nmid j$ let s_j be the least integer with $jp^{s_j} > \ell$.

Proposition 3.1. *$\mathcal{G}_\ell$ is the internal direct product of the cyclic groups generated by $1 + \omega_i u^j$, $p \nmid j \le \ell$, $0 \le i < k$, and $1 + \omega_i u^j$ has order p^{s_j}. In particular $\mathcal{G}_\ell \cong \prod_{p \nmid j \le \ell} (\mathbb{Z}/p^{s_j})^k$.*

Proof. In characteristic p, $(1 + \omega u^j)^{p^s} = 1 + \omega^{p^s} u^{jp^s}$, which is 1 modulo $u^{\ell+1}$ exactly when $jp^s > \ell$; this gives the orders. The orders multiply to $\prod_j q^{s_j} = q^{\#\{(j,s)\,:\, jp^s \le \ell\}} = q^\ell$, because every $m \le \ell$ is uniquely jp^s with $p \nmid j$. It remains to show that the elements generate. Let $x = 1 + c_m u^m + O(u^{m+1})$ with $c_m \ne 0$, write $m = jp^a$, and expand $c_m^{p^{-a}} = \sum_i e_i \omega_i$ with $e_i \in \mathbb{F}_p$. Then $\prod_i (1 + \omega_i u^j)^{e_i p^a} = 1 + \sum_i e_i \omega_i^{p^a} u^m + O(u^{m+1}) = 1 + c_m u^m + O(u^{m+1})$, so dividing x by this product raises the order of vanishing of $x - 1$. Induction on m expresses x as a product of the proposed generators. □

The proof is an algorithm: it computes the coordinates of $[g]$ (its *discrete logarithm*) in $O(\ell^2 p)$ operations in $\mathbb{F}_q$. Characters of $\mathcal{G}$ are then $\chi_b(x) = \prod_{(j,i)} \zeta_{p^{s_j}}^{-a_{j,i}(x) b_{j,i}}$, and sums over $\mathcal{G}$ weighted by characters become multidimensional discrete Fourier transforms with axis lengths p^{s_j}.

4 The algorithm

1. For $k = 0, \dots, \ell - 1$ form the indicator of $\{[g] : \deg g = k\}$ in the coordinates of Proposition 3.1 (for $k \le \ell$ distinct g give distinct classes), and transform it to obtain $c_k(\chi)$ for all χ.

2. For every χ run (1) up to $n = 2d$.
3. Set $\psi_{2d}(1) = q^{2d}$ and invert the transform to obtain $\Psi(x)$ for all x by (2).

Step 1 costs ℓ transforms, step 2 costs $O(d^2)$ per character, step 3 one transform. With naive transforms along each axis the total is $O(d^2 q^\ell + d\,\sigma\, q^\ell)$ operations, where $\sigma = k \sum_{p \nmid j \le \ell} p^{s_j}$ is the sum of the orders of the $k \cdot \#\{j \le \ell : p \nmid j\}$ cyclic factors in Proposition 3.1. Testing every polynomial costs about q^{2d} instead.

Exact arithmetic. All quantities in (1)–(2) lie in $\mathbb{Z}[\zeta][1/q]$, where ζ is a primitive p^S-th root of unity and $p^S = \max_j p^{s_j}$. For a prime $M \equiv 1 \pmod{p^S}$, $M \neq p$, choose $z \in \mathbb{F}_M$ of order p^S; the ring map $\zeta \mapsto z$ turns (1)–(2) into identities in $\mathbb{F}_M$. Running the algorithm in $\mathbb{F}_M$ therefore returns $\Psi(x) \bmod M$ exactly. Since $0 \le \Psi(x) \le 2d\,q^{d+1}$, two primes $M_1, M_2 < 2^{31}$ with $M_1 M_2 > 2dq^{d+1}$ determine $\Psi(x)$ by the Chinese remainder theorem. All products fit in 64-bit integers.

Checks. A floating-point version (fast Fourier transforms) is used for exploration. The exact and floating-point results agree on every class tested; both agree with brute-force counts for all 11,232 orbit representatives computed in [6]; and $\sum_x \Psi(x) = q^{2d}$ holds exactly in every run. For example $q = 5$, $d = 8$ took one hour by brute force in [6] and 0.4 s here.

5 Verification

Recall that $E(f) = \sum_{P \in \mathcal{I}_f,\ P \text{ not irreducible}} \Lambda(P)$ is the contribution of the proper prime powers, so that $2d\,N_{\mathrm{irr}}(f) = \Psi(f) - E(f)$. Put $a = 1$ if q is odd and $a = \lceil (d+1)/2 \rceil$ if q is even.

Lemma 5.1. *For every monic f of degree d, $E(f) \le d\,q^a + 2q^{\lfloor 2d/3 \rfloor}$.*

Proof. A proper prime power in $\mathcal{I}_f$ is $P = \pi^j$ with π monic irreducible, $j \ge 2$, $j \mid 2d$, and $\Lambda(P) = \deg \pi = 2d/j$.

Squares ($j = 2$). Here $\deg \pi = d$ and $\pi^2 - f^2 = s$ has degree $\le d$. If q is odd, $\pi^2 - f^2 = (\pi - f)(\pi + f)$ with $\deg(\pi + f) = d$, so $\pi - f$ is a constant: at most q such π. If q is even, $\pi^2 - f^2 = (\pi - f)^2$, so $\deg(\pi - f) \le d/2$: at most $q^{\lfloor d/2 \rfloor + 1} = q^{\lceil (d+1)/2 \rceil}$ such π. Each contributes d, giving at most $d\,q^a$. (The exponent a is the reason the bound is weaker in characteristic 2: π^2 has no odd-degree terms, so the prescribed coefficients in odd positions carry no information about π.)

Higher powers ($j \ge 3$). Then $m = \deg \pi \le 2d/3$. Summing over *all* monic irreducibles of degree $m \le \lfloor 2d/3 \rfloor$, whether or not $\pi^j \in \mathcal{I}_f$, and using $m\,I_m(q) = \sum_{e \mid m} \mu(m/e) q^e \le q^m$ for the number $I_m(q)$ of monic irreducibles of degree m (μ the Möbius function), the contribution is at most $\sum_{m \le \lfloor 2d/3 \rfloor} q^m < q^{\lfloor 2d/3 \rfloor + 1}/(q-1) \le 2q^{\lfloor 2d/3 \rfloor}$. □

An interval is *certified* if $\Psi(f) > d\,q^a + 2q^{\lfloor 2d/3 \rfloor}$. By Lemma 5.1, $2d\,N_{\mathrm{irr}}(f) = \Psi(f) - E(f) > 0$ for a certified interval, so it contains an irreducible polynomial. The bound of Lemma 5.1 also recovers the classical range from the Weil bound alone.

Proposition 5.2 (the range $q \ge d - 1$)**.** *For every monic f of degree $d \ge 2$, $|\Psi(f) - q^{d+1}| \le (d-2)q^d$. If $q \ge d - 1$ and $q \ge 4$, every Legendre interval is certified; in particular it contains an irreducible polynomial.*

Proof. By Proposition 7.1 (proved in Section 7, independently of this section), $|\psi_{2d}(\chi)| \le (m(\chi) - 2)q^d$ for $\chi \neq 1$, and $\sum_{\chi \neq 1}(m(\chi) - 2) = \sum_{m=3}^{\ell+1}(m-2)(q^{m-1} - q^{m-2}) = (\ell - 1)q^\ell$ by telescoping. Inserting this in (2) gives the first claim. Hence $\Psi(f) \ge q^d(q - d + 2)$, which is $\ge q^d$ when $q \ge d - 1$, and it remains to compare with $R = d\,q^a + 2q^{\lfloor 2d/3 \rfloor}$. For $d = 2$ the

Weil term is absent, $\Psi(f) = q^3$, and $R \le 2q^2 + 2q < q^3$ for $q \ge 3$. For $d \ge 3$ and odd $q \ge 5$: $\lfloor 2d/3 \rfloor \le d-1$ and $d \le q+1$, so $q^d - R \ge q^{d-1}(q-2) - (q+1)q \ge 3q^2 - q^2 - q > 0$. For $d \ge 5$ and even $q \ge 4$: $a \le d-2$ and $\lfloor 2d/3 \rfloor \le d-2$, so $R \le (d+2)q^{d-2} \le (q+3)q^{d-2} < q^d$. For $d \in \{3,4\}$ and even $q \ge 4$ we use the full lower bound $\Psi(f) \ge q^{d+1} - (d-2)q^d$: $q^4 - q^3 > 5q^2 = R$ for $d = 3$, and $q^5 - 2q^4 > 4q^3 + 2q^2 = R$ for $d = 4$, both for $q \ge 4$. □

Theorem 5.3 (computer-assisted)**.** *Every Legendre interval over $\mathbb{F}_q$ contains an irreducible polynomial in each of the following cases: $q = 2$, $d \le 25$; $q = 3$, $d \le 16$; $q = 4$, $d \le 13$; $q = 5$, $d \le 11$; $q = 7$, $d \le 10$; $q = 8$, $d \le 10$; $q = 9$, $d \le 8$. Consequently the function field analogue of Legendre's conjecture holds for every prime power q when $d \le 10$.*

Proof. For each listed (q, d) the exact algorithm computes Ψ on every Legendre class, and all classes are certified except some classes in the cases $(2,2)$, $(2,3)$, $(2,4)$, $(2,6)$. For these four cases Table 1 lists every Legendre interval with $\Psi(f)$, $E(f)$ and $N_{\mathrm{irr}}(f)$, obtained by testing every element and enumerating the prime powers; in each of them $N_{\mathrm{irr}}(f) \ge 1$. For the second statement let $d \le 10$. If $q \ge \max(d-1, 4)$, Proposition 5.2 applies. Otherwise $q \le 3$ or $q \le d-2 \le 10-2$, so $q \in \{2,3,4,5,7,8\}$, and for each of these $d \le 10$ lies within the verified range. □

q	d	f	$\Psi(f)$	$E(f)$	$N_{\mathrm{irr}}(f)$	bound
2	2	t^2	8	4	1	12
2	3	t^3	16	4	2	20
2	3	$t^3 + t^2$	16	4	2	20
2	4	t^4	24	8	2	40
2	4	$t^4 + t^3$	24	8	2	40
2	6	t^6	160	16	12	128
2	6	$t^6 + t^4$	160	16	12	128
2	6	$t^6 + t^5$	164	20	12	128
2	6	$t^6 + t^5 + t^4$	60	12	4	128

Table 1: All Legendre intervals for the four cases in which some interval is not certified by the bound (last column). Each row is one interval, given by a representative f (with $f_0 = 0$). Source: `data/low_degree_char2.csv`, produced by `code/low_degree_cases.py`.

The case $d = 11$ would require $q = 7$, 8 and 9 with $d = 11$, that is up to $9^{10} \approx 3.5 \times 10^9$ classes, beyond the memory used here.

6 Fluctuations across intervals

For odd q every class is a Legendre interval (Lemma 2.1), so the distribution of Ψ over Legendre intervals is the distribution over all q^{d-1} classes. Keating and Rudnick [13] proved $\operatorname{Var}\Psi \sim (d-2)q^{d+1}$ as $q \to \infty$ (for $d \ge 4$).

For odd q the ratio $\operatorname{Var}\Psi/((d-2)q^{d+1})$ is between 0.96 and 0.99 for the largest computed d of each q (Figure 1), and Z is close to Gaussian (Figure 2). The minimum of Ψ approaches q^{d+1} as d grows, as the Gaussian heuristic of [6] predicts.

7 The zeros

Proposition 7.1. *For $\chi \ne 1$ let $m(\chi)$ be the least m such that χ is trivial on $\{x \in \mathcal{G} : x \equiv 1 \bmod u^m\}$, so $2 \le m(\chi) \le \ell + 1$. Then $L(z,\chi)$ has degree exactly $m(\chi) - 2$, and all its inverse*

q	d	intervals	$\min \Psi/q^{d+1}$	certified	open range	time (s)
2	23	2,048	0.9961	2,048	yes	41.1
2	24	2,048	0.9964	2,048	yes	87.7
2	25	4,096	0.9983	4,096	yes	184.6
3	14	1,594,323	0.9958	1,594,323	yes	7.6
3	15	4,782,969	0.9971	4,782,969	yes	25.1
3	16	14,348,907	0.9982	14,348,907	yes	81.7
4	11	1,024	0.9981	1,024	yes	2.6
4	12	1,024	0.9968	1,024	yes	11.8
4	13	4,096	0.9995	4,096	yes	53.7
5	9	390,625	0.9961	390,625	yes	1.1
5	10	1,953,125	0.9977	1,953,125	yes	5.8
5	11	9,765,625	0.9991	9,765,625	yes	34.1
7	8	823,543	0.9982	823,543	no	3.4
7	9	5,764,801	0.9993	5,764,801	yes	26.7
7	10	40,353,607	0.9997	40,353,607	yes	212.4
8	8	512	0.9963	512	no	2.5
8	9	4,096	0.9993	4,096	no	34.1
8	10	4,096	0.9999	4,096	yes	338.6
9	6	59,049	0.9889	59,049	no	0.1
9	7	531,441	0.9987	531,441	no	0.6
9	8	4,782,969	0.9994	4,782,969	no	6.0

Table 2: Exact verification. "Intervals" is the number of Legendre classes; the ratio is $\min_f \Psi(f)/q^{d+1}$; times are wall-clock for the exact computation (two primes). Only the largest d for each q and the cases with $q \le d-2$ near the boundary are shown; all cases are in `data/exact_scan.csv`.

roots have absolute value $q^{1/2}$. Exactly $q^{m-1} - q^{m-2}$ characters have $m(\chi) = m$.

Proof. Write $\hat{g}(u) = u^{\deg g} g(1/u)$. For monic g of degree k, $\hat{g}$ runs over the polynomials h with $h(0) = 1$ and $\deg h \le k$, bijectively, and $[g] = h \bmod u^{\ell+1}$. Hence

$$L(z,\chi) = \sum_{h(0)=1} \chi(h)\, z^{\deg h} \sum_{k \ge \deg h} z^k = \frac{L_{\mathrm{D}}(z,\chi)}{1-z}, \qquad L_{\mathrm{D}}(z,\chi) = \sum_{h(0)=1} \chi(h)\, z^{\deg h},$$

where $L_{\mathrm{D}}(z,\chi)$ is the Dirichlet L-function of χ viewed as an even character modulo $u^{\ell+1}$ in $\mathbb{F}_q[u]$ (scaling by the constant term identifies polynomials with $h(0) = 1$ with monic polynomials prime to u, and χ is trivial on constants). The character is induced from a primitive even character modulo $u^{m(\chi)}$, and since u is the only prime dividing either modulus, L_{D} is unchanged by passing to the primitive character. For a primitive even character modulo a polynomial of degree $m \ge 2$, $L_{\mathrm{D}}(z) = (1-z)P(z)$ with P of degree $m-2$ whose inverse roots all have absolute value $q^{1/2}$ (Weil; see [15, Ch. 4] and [13, §4]). So $L(z,\chi) = P(z)$. The characters with $m(\chi) \le m$ are those of the quotient $\mathcal{G}_{m-1}$, which has q^{m-1} elements. □

The computed L-functions confirm this exactly: in every case of Table 3 the number of characters of each conductor u^m is $q^{m-1} - q^{m-2}$, the coefficient $c_{m-2}(\chi)$ is nonzero and all higher ones vanish. The inverse roots were then computed numerically from the exact-degree polynomials. Their absolute values are $\sqrt{q}$ to within 10^{-6} relative error, except for roots of multiplicity greater than one, where floating-point root finding is less accurate; the largest

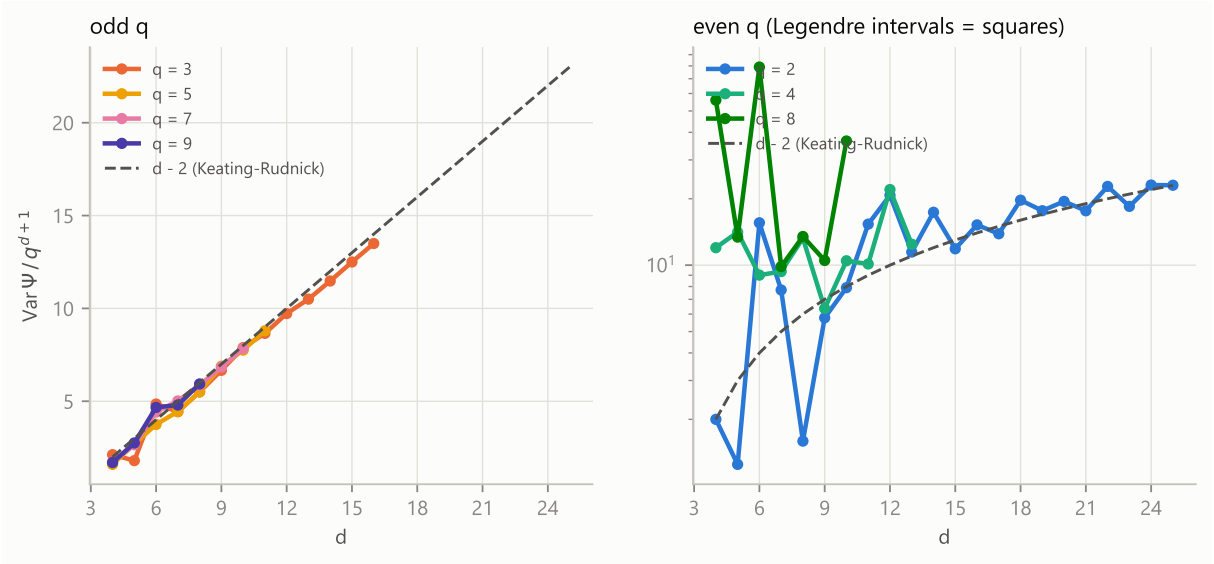


Figure 1: $\mathrm{Var}_f\,\Psi(f)/q^{d+1}$ (mean square deviation from q^{d+1}) over all Legendre intervals. Left: odd q, against the Keating–Rudnick limit $d-2$. Right: even q, where the prediction fails.

deviation is 1.0×10^{-4}. The only such case with a deviation above 10^{-5} is one quadratic character for $(q,d)=(4,10)$, whose L-function is $(1+2z)^4(1-2z+4z^2)^2$.

Let $F(n)=(q^\ell-1)^{-1}\sum_{\chi\neq1}|\psi_n(\chi)|^2/q^n$. Writing the inverse roots as $\sqrt{q}\,e^{i\theta}$, $F(n)$ is the average of $|\sum e^{in\theta}|^2$; for $N\times N$ CUE matrices this average is $\min(n,N)$.

Proposition 7.2. *For $1\le n\le\ell$,*

$$F(n)=\frac{q^\ell S_n-q^{2n}}{(q^\ell-1)\,q^n},\qquad S_n=\sum_{\deg g=n}\Lambda(g)^2=\sum_{e\mid n}e^2I_e(q),$$

where $I_e(q)$ is the number of monic irreducible polynomials of degree e. In particular $F(n)=n+O(n^2q^{-n/2})+O(q^{n-\ell})$, and $F(2)=2-1/q+O(q^{2-\ell})$.

Proof. By Parseval, $\sum_\chi|\psi_n(\chi)|^2=q^\ell\sum_x\big(\sum_{\deg g=n,[g]=x}\Lambda(g)\big)^2$. For monic $g=t^n+g_{n-1}t^{n-1}+\cdots+g_0$ we have $[g]=1+g_{n-1}u+\cdots+g_0u^n$ modulo $u^{\ell+1}$. If $n\le\ell$ no coefficient is lost in the truncation, so $g\mapsto[g]$ is injective on monic polynomials of degree n: exactly q^n classes contain one such polynomial and the other $q^\ell-q^n$ contain none. The inner sum therefore has at most one term and the total is $q^\ell S_n$. Remove the trivial character, $|\psi_n(1)|^2=q^{2n}$. □

The computed $F(n)$ agree with Proposition 7.2 to 10^{-14}. For $n>\ell$ there is no such formula; there $F(n)$ is the variance of prime counts in classes of degree-n polynomials, and it stays within 9% of the CUE value $\ell-1$ (Figure 3). The nearest-neighbour spacings of the eigenangles agree with CUE$(d-2)$ to within 0.05 in density.

8 Characteristic 2

8.1 Structure

Let q be even. By Lemma 2.1 the Legendre intervals form the subgroup $\mathcal{G}^2$ of index $2^{k\cdot\#\{j\le\ell\text{ odd}\}}$. Their elements have a special form.

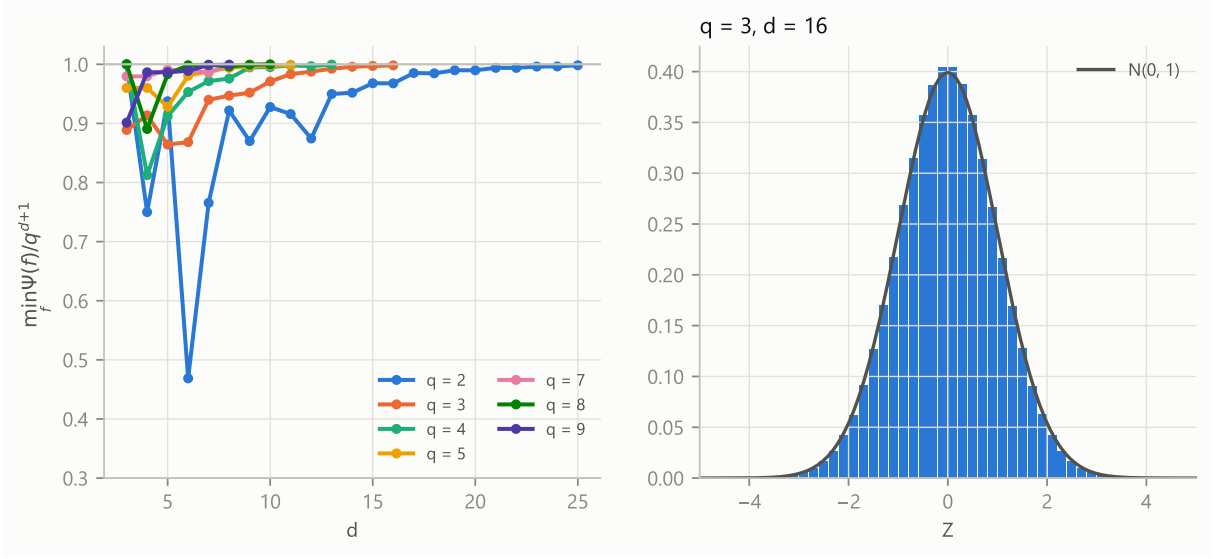


Figure 2: Left: $\min_f \Psi(f)/q^{d+1}$ against d. Right: the distribution of $Z = (\Psi - q^{d+1})/\sqrt{(d-2)q^{d+1}}$ over the 14,348,907 intervals with $(q,d) = (3,16)$, against the standard normal density.

q	d	characters	generic	exact	conductor counts	max dev. of $\lvert\omega\rvert/\sqrt{q}$
2	20	400,000	0.5009	0.5000	as predicted	4.8e-06
3	12	177,146	0.6667	0.6667	as predicted	8.9e-06
3	14	400,000	0.6672	0.6667	as predicted	7.0e-06
4	10	262,143	0.7500	0.7500	as predicted	1.0e-04
5	9	390,624	0.8000	0.8000	as predicted	3.0e-08
7	8	400,000	0.8576	0.8571	as predicted	1.7e-08

Table 3: Zeros of the Hayes L-functions for the listed (q,d) (all characters, or a random sample of 400,000). "Generic" is the proportion of characters with conductor $u^{\ell+1}$, i.e. with the maximal degree $d-2$; the exact proportion is $(q^\ell - q^{\ell-1})/(q^\ell - 1)$. The last column is the largest relative deviation of $|\omega|$ from $\sqrt{q}$ over all computed inverse roots.

Proposition 8.1. *Let q be even and f monic of degree d. Every $P \in \mathcal{I}_f$ can be written uniquely as*

$$P = h^2 + t\,b^2, \qquad h = f + a,\ \deg a \le \lfloor d/2 \rfloor,\ \deg b \le \lfloor (d-1)/2 \rfloor.$$

Then $P' = b^2$, and P is squarefree if and only if $\gcd(h,b) = 1$.

Proof. Write $s = P - f^2$ as $a(t)^2 + t\,b(t)^2$ by separating even and odd powers (every element of $\mathbb{F}_q$ is a square); then $f^2 + s = (f+a)^2 + tb^2$ because squaring is additive. Differentiating, $P' = b^2$. If $\pi^2 \mid P$ for an irreducible π, then $\pi \mid P' = b^2$, so $\pi \mid b$ and $\pi \mid h^2$, so $\pi \mid h$. Conversely if $\pi \mid h$ and $\pi \mid b$ then π^2 divides both h^2 and tb^2. If $b = 0$ then $P = h^2$ is not squarefree, and $\gcd(h,0) = h$ is nonconstant. □

Thus a Legendre interval in characteristic 2 is a union, over the $q^{\lfloor (d+1)/2 \rfloor}$ values of b, of the values of the quadratic polynomial $X^2 + tb^2$ on a short interval of X around f. Counting its primes is closer to a Bateman–Horn problem for quadratic polynomials than to counting primes in an interval.

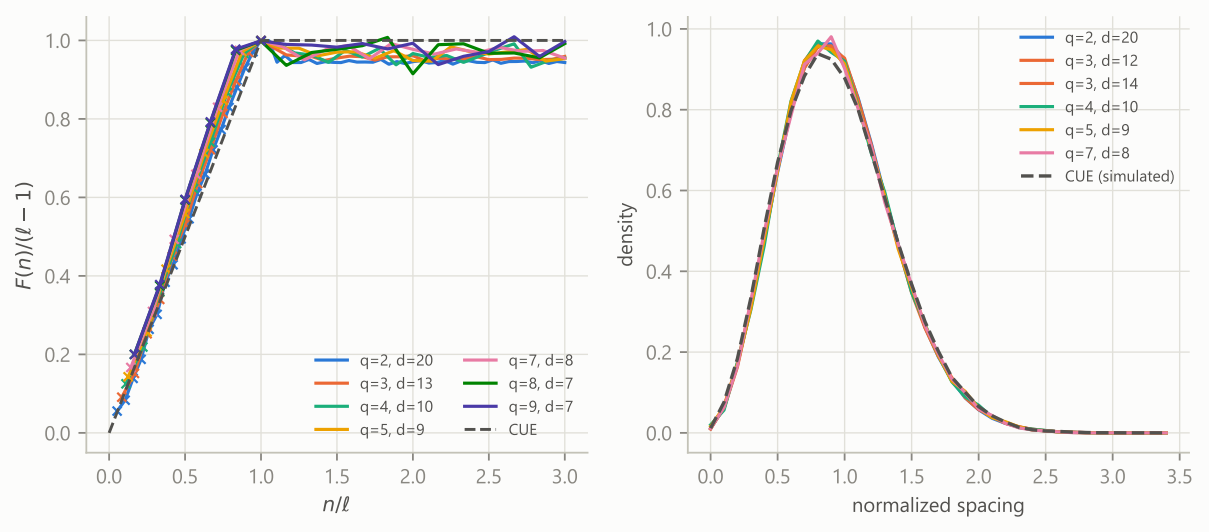


Figure 3: Left: the form factor $F(n)/(\ell-1)$ against n/ℓ for several (q,d), with the CUE value $\min(n,\ell-1)/(\ell-1)$ and the exact diagonal of Proposition 7.2 (crosses). Right: nearest-neighbour spacing densities against CUE.

8.2 The anomaly

Table 4 compares the variance of Ψ over all classes of $\mathcal{G}$ with the variance over the Legendre intervals $\mathcal{G}^2$. Over all classes the variance follows $d-2$ for even q just as for odd q (cf. [12], where the Frobenius equidistribution for the characters of $\mathcal{G}$ is obtained through Witt vectors in arbitrary characteristic), and the form factor over all characters matches CUE for $q=2,4,8$ (Figure 3). Over $\mathcal{G}^2$ the variance differs from it by a factor that ranges from about 0.33 to 21, while the mean stays close to q^{d+1}. The anomaly is therefore a property of the subfamily $\mathcal{G}^2$, not of the L-functions as a whole.

By Parseval on $\mathcal{G}^2$, $\mathrm{Var}_{\mathcal{G}^2}\Psi$ is governed by the sums $\sum_\varepsilon \psi_{2d}(\chi\varepsilon)$ over characters ε trivial on $\mathcal{G}^2$, i.e. the characters of order ≤ 2. If $\psi_{2d}(\chi)$ and $\psi_{2d}(\chi\varepsilon)$ were uncorrelated, the variance would equal the full-group value. Define

$$\rho(\varepsilon)=\frac{\mathrm{Re}\sum_{\chi\neq 1}\psi_{2d}(\chi)\overline{\psi_{2d}(\chi\varepsilon)}}{\sum_{\chi\neq 1}|\psi_{2d}(\chi)|^2}. \tag{3}$$

The ratio of the two variances is $1+\sum_{\varepsilon\neq 1}\rho(\varepsilon)$ up to the trivial-character terms, and in every case of Table 4 the two columns coincide to four decimals. The whole anomaly is thus carried by correlations between quadratic twists.

Where the correlations sit. Let q be even. By Proposition 3.1, $\mathcal{G}\cong\prod_{j\le\ell\ \mathrm{odd}}(\mathbb{Z}/2^{s_j})^k$, and $\mathcal{G}^2$ is the subgroup of elements whose coordinates are all even. Hence $\mathcal{G}/\mathcal{G}^2\cong(\mathbb{Z}/2)^{km}$ with $m=\#\{j\le\ell\ \text{odd}\}$, and the characters trivial on $\mathcal{G}^2$ are $\varepsilon(x)=(-1)^{\sum_{(j,i)}e_{j,i}a_{j,i}(x)}$ with $e_{j,i}\in\{0,1\}$. The coordinates with $j>\ell/2$ are exactly those with $s_j=1$, that is, the generators $1+\omega_i u^j$ of order 2; there are $k\cdot\#\{j\ \text{odd}:\ \ell/2<j\le\ell\}$ of them. One might expect the twists supported on these coordinates to carry the effect, but they do not: Table 4 splits $\sum_{\varepsilon\neq1}\rho(\varepsilon)$ into the part from ε with $e_{j,i}=0$ for all $j<\ell/2$ and the rest, and in most cases the rest dominates (for example 1.67 against 18.67 for $(q,d)=(8,4)$). The correlations are spread over many quadratic twists, and we have no structural description of which twists correlate.

Two natural explanations fail. On fixed-b slices, $\mu(h^2+tb^2)$ is neither of the form $\pm(-1)^{L(a)}$ with L affine over $\mathbb{F}_2$ in the coefficients of a, nor a function of $h \bmod b$ alone (tested for $d=$

q	d	mean	Var (all)	Var (Legendre)	ratio	excess	$\Sigma\rho$, order 2	$\Sigma\rho$, other
2	4	0.7500	1.50	2.00	1.33	1.33	-0.33	+0.67
2	5	1.0625	1.17	1.25	1.07	1.07	-0.36	+0.43
2	6	1.0625	5.76	15.56	2.70	2.70	+0.53	+1.18
2	7	1.0469	3.88	7.72	1.99	1.99	+0.24	+0.74
2	8	0.9688	4.88	1.59	0.33	0.33	+0.21	-0.88
2	9	0.9805	7.10	5.77	0.81	0.81	-0.40	+0.21
2	10	1.0273	7.19	7.91	1.10	1.10	+0.24	-0.14
2	11	0.9985	8.57	15.35	1.79	1.79	+0.21	+0.58
2	12	0.9839	9.75	20.74	2.13	2.13	+0.13	+1.00
2	13	1.0044	10.10	11.47	1.14	1.14	-0.09	+0.23
2	14	0.9995	11.30	17.37	1.54	1.54	+0.24	+0.29
2	15	0.9995	12.10	11.86	0.98	0.98	+0.02	-0.04
2	16	0.9983	13.19	15.22	1.15	1.15	+0.04	+0.12
4	4	1.0000	2.25	12.00	5.33	5.33	+0.33	+4.00
4	5	1.0117	2.60	14.11	5.43	5.43	+0.83	+3.60
4	6	0.9883	3.70	9.01	2.44	2.44	+1.34	+0.09
4	7	1.0037	5.23	9.36	1.79	1.79	+0.42	+0.36
4	8	1.0000	5.88	13.25	2.25	2.25	+0.07	+1.18
4	9	0.9999	6.79	6.36	0.94	0.94	-0.02	-0.04
8	4	0.9727	2.62	56.00	21.33	21.33	+1.67	+18.67
8	5	1.0017	3.11	13.41	4.31	4.31	+0.28	+3.03
8	6	1.0017	4.75	79.09	16.66	16.66	+2.21	+13.45
8	7	1.0002	4.96	9.85	1.99	1.99	+0.00	+0.98

Table 4: Characteristic 2. The two variance columns are the mean of $(\Psi - q^{d+1})^2/q^{d+1}$ over all classes of $\mathcal{G}$ and over the Legendre intervals $\mathcal{G}^2$; "mean" is the mean of Ψ/q^{d+1} over $\mathcal{G}^2$; "ratio" is the quotient of the two variances, and "excess" is $1 + \sum_{\varepsilon \neq 1} \rho(\varepsilon)$ from (3). Source: `data/char2_variance.csv`.

$8, 10, 12$, $q = 2$).

The natural framework is the parity of the number of prime factors. In odd characteristic it is given by Pellet's formula through the discriminant; in characteristic 2 Pellet's formula fails, and Carmon [2] replaced the discriminant by Berlekamp's discriminant to prove the function field Chowla conjecture in characteristic 2 in the limit $q \to \infty$. For $P = h^2 + tb^2$ the derivative $P' = b^2$ depends only on b, which is the situation in which Sawin and Shusterman [17] turn the Möbius function into a Dirichlet character in odd characteristic. Whether an analogue of their mechanism, through Berlekamp's discriminant, accounts for the correlations is open.

Open problem 8.2. *Explain the excess variance of Ψ over Legendre intervals in characteristic 2: determine $\lim \mathrm{Var}_{\mathcal{G}^2} \Psi/q^{d+1}$, as $q \to \infty$ through powers of 2 and as $d \to \infty$, and identify the mechanism behind the correlations $\rho(\varepsilon)$, presumably through the Berlekamp discriminant and the form $P = h^2 + tb^2$ with $P' = b^2$ [2, 17].*

Note added. In the sequel [3], Problem 8.2 is resolved for $d = 4$: $\mathrm{Var}_{\mathcal{G}^2} \Psi/q^5 = q(q-1)$ and $\mathrm{Var}_{\mathcal{G}} \Psi/q^5 = 3 - 3/q$ are proved for all $q = 2^k$ via the quadratic tower $\mathbb{F}_q \subset \mathbb{F}_{q^2} \subset \mathbb{F}_{q^4} \subset \mathbb{F}_{q^8}$, and the correlations $\rho(\varepsilon)$ are determined ($\rho = 1$ on the $q-1$ twists of conductor u^2, $\rho = (q-3)/(3(q-1))$ on all others for odd k). For $d = 6$ the Legendre profile takes exactly three values, governed by three constants A_6, X, Y; A_6 is evaluated for all odd k, and the variance satisfies $\mathrm{Var}_{\mathcal{G}^2} \Psi/q^7 \asymp q^2$, excluding the q^4 law one might have expected. The constant X is treated in Parts III and IV [4, 5], where the relevant character sums of order 4 are governed by the symplectic group $Sp_4(\mathbb{F}_2) \cong S_6$ acting on a hexad of roots of an explicit sextic. The

mechanism found there is the additive structure of the subgroup $\mathcal{G}^2$ and the quadratic towers $\mathbb{F}_{q^{2d}}/\mathbb{F}_{q^d}$, rather than the Berlekamp discriminant suggested above.

9 Discussion

The explicit formula reduces the verification of the function field Legendre conjecture to a transform over the Hayes group, and exact modular arithmetic makes the result a proof. The method reaches every q for $d \le 10$; going further needs memory proportional to q^{d-1}, so the next step, $q = 7, 8, 9$ with $d = 11$, needs either a distributed transform or a reduction by symmetry. The substitutions $t \mapsto \lambda t + b$ and the Frobenius automorphism of $\mathbb{F}_q$ act on classes of degree $2d$ by affine maps of $\mathcal{G}$ (an automorphism followed by a translation) and preserve Ψ; for $q = 9$ this group has order $2 \cdot 9 \cdot 8 = 144$. A transform adapted to this action could reduce the memory by a factor of that order, but we have not implemented it, and whether it brings $(9, 11)$ within reach of one machine is untested.

The data support the conjecture of [6] that no Legendre interval is ever empty. For odd q the fluctuations behave as random matrix theory predicts already at $q = 3$; in characteristic 2 they do not, for a reason that Problem 8.2 asks to explain.

Code and data. All programs and results are in the repository

`https://github.com/Ruqing1963/legendre-intervals-explicit-formula`

The files named in the text (`exact_scan.csv`, `char2_variance.csv`, `low_degree_char2.csv`) are in its `data` directory. This paper is archived as doi:10.5281/zenodo.23225896.

References

[1] E. Bank, L. Bary-Soroker, L. Rosenzweig, Prime polynomials in short intervals and in arithmetic progressions, *Duke Math. J.* **164** (2015), 277–295.

[2] D. Carmon, The autocorrelation of the Möbius function and Chowla's conjecture for the rational function field in characteristic 2, *Philos. Trans. Roy. Soc. A* **373** (2015). arXiv:1409.3694.

[3] R. Chen, Prime polynomials in Legendre intervals over $\mathbb{F}_{2^k}[t]$, II: exact variances via quadratic towers and supersingular Artin–Schreier curves, preprint (2026), DOI 10.5281/zenodo.23252762.

[4] R. Chen, Prime polynomials in Legendre intervals over $\mathbb{F}_{2^k}[t]$, III: the hexad of a quartic linearised polynomial, the symplectic group $Sp_4(\mathbb{F}_2)$, and the Witt sums behind the constant X, preprint (2026), DOI 10.5281/zenodo.23290724.

[5] R. Chen, Prime polynomials in Legendre intervals over $\mathbb{F}_{2^k}[t]$, IV: point counts on hexad modular curves, Brown invariants of Witt sheaves, and the constant X, preprint (2026), DOI 10.5281/zenodo.23308682.

[6] R. Chen, Legendre's conjecture in function fields: the classical range, full monodromy, and the open range $q \le d-2$ (v7), Zenodo, 2026. `https://doi.org/10.5281/zenodo.23185896`

[7] S. D. Cohen, Explicit theorems on generator polynomials, *Finite Fields Appl.* **11** (2005), 337–357.

[8] Z. Gao, Improved error bounds for the number of irreducible polynomials and self-reciprocal irreducible monic polynomials with prescribed coefficients over a finite field, arXiv:2109.14154.

[9] J. Ha, Irreducible polynomials with several prescribed coefficients, *Finite Fields Appl.* **40** (2016), 95–121.

[10] D. R. Hayes, The distribution of irreducibles in $GF[q, x]$, *Trans. Amer. Math. Soc.* **117** (1965), 101–127.

[11] C. N. Hsu, The distribution of irreducible polynomials in $\mathbb{F}_q[t]$, *J. Number Theory* **61** (1996), 85–96.

[12] N. M. Katz, Witt vectors and a question of Keating and Rudnick, *Int. Math. Res. Not. IMRN* 2013, no. 16, 3613–3638.

[13] J. P. Keating, Z. Rudnick, The variance of the number of prime polynomials in short intervals and in residue classes, *Int. Math. Res. Not. IMRN* 2014, no. 1, 259–288.

[14] P. Pollack, Irreducible polynomials with several prescribed coefficients, *Finite Fields Appl.* **22** (2013), 70–78.

[15] M. Rosen, *Number Theory in Function Fields*, Graduate Texts in Mathematics 210, Springer, 2002.

[16] Z. Rudnick, Some problems in analytic number theory for polynomials over a finite field, in: *Proceedings of the International Congress of Mathematicians (Seoul, 2014)*, Vol. II, Kyung Moon Sa, Seoul, 2014, 443–459.

[17] W. Sawin, M. Shusterman, On the Chowla and twin primes conjectures over $\mathbb{F}_q[T]$, *Ann. of Math.* **196** (2022). arXiv:1808.04001.

[18] D. Wan, Generators and irreducible polynomials over finite fields, *Math. Comp.* **66** (1997), 1195–1212.